Carelixa: An AI-Driven Collaborative Platform for Autism Spectrum Disorder Progress Monitoring

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Abstract: Effective support for children with Autism Spectrum Disorder (ASD) requires steady cooperation between parents, teachers, and clinicians. In practice, however, these three groups work in isolation. A child may appear calm and engaged during structured classroom routines but experience sensory meltdowns at home. When clinicians only see a child during brief visits spaced months apart, subtle developmental changes are easily missed. This paper presents Carelixa, a software platform built to connect these stakeholders and track progress over time. Parents, teachers, and clinicians record weekly functional observations alongside standardized M-CHAT-R/F screening. We analyzed 600 weekly records from 50 children to train machine learning models that predict whether a child's developmental trajectory will improve or decline in the following week. After dropping the unlagged first week for each child, 550 transition instances were evaluated using an 80/20 train/test split (440 training, 110 testing). The proposed multi-observer Random Forest classifier reached a test accuracy of 64.55% and a 5-fold cross-validation accuracy of 67.27% (peak fold 77.27%), outperforming single-observer baselines relying only on parent reports (62.73% test / 65.45% CV) or teacher reports (63.64% test / 66.73% CV), as well as standalone decision trees (60.00%) and support vector machines (54.55%). The system also includes an LLM module that turns identified skill weaknesses into classroom adjustments and home activities, along with an Ethereum smart contract for parental consent management. These findings show that combining observations from multiple environments gives a more reliable picture of pediatric autism progress than relying on any single observer.

Keywords: Autism Spectrum Disorder, Progress Tracking, Machine Learning, Multi-Observer Telemetry, Clinical Decision Support.

Introduction: Autism Spectrum Disorder (ASD) involves differences in social communication, sensory reactivity, and repetitive behavioral patterns (Lord et al., 2018). While clinical guidance calls for regular, cross-setting intervention (World Health Organization, 2022), care delivery in practice is fragmented. Parents observe home behavior, sleep issues, and evening meltdowns; educators observe classroom participation, peer engagement, and task transitions; clinicians conduct diagnostic evaluations during scheduled visits. Because these observations are recorded independently, therapists frequently make decisions based on retrospective recall during consultations scheduled three to six months apart (Robins et al., 2014). This creates a known masking problem: a child may expend tremendous effort complying with classroom rules but experience sensory overload at home. An isolated teacher assessment reports progress, while an isolated parent report indicates deterioration. Furthermore, early regressions in speech or emotional self-regulation can persist for weeks before anyone notices. Carelixa was built to unify these observation streams, identify behavioral discrepancies between settings, predict trajectory shifts, and supply automated, practical support strategies.

Objective (s): The primary objectives of this study are: (1) to develop a multi-role web platform where parents, teachers, and clinicians record weekly functional and qualitative metrics; (2) to train a machine learning model that forecasts short-term trajectory shifts using tri-party inputs and cross-setting score gaps; (3) to evaluate whether multi-observer models achieve higher predictive accuracy than single-observer baselines; (4) to generate tailored weekly recommendations using a large language model based on identified skill weaknesses; and (5) to implement an on-chain smart contract protocol that grants parents direct sovereignty over pediatric data access.

Methodology: Carelixa contains four connected components:

1. Multi-Observer Data Ingestion: Role-specific web dashboards allow parents, teachers, and clinicians to submit weekly Likert ratings (1 to 5) across four domains: Communication (Q1, Q2), Social Interaction (Q3, Q4), Emotional Regulation (Q5, Q6), and Home Daily Living (P1, P2, P3). Parents also log meltdown counts and sleep quality; teachers track classroom participation; clinicians record milestone reviews and digitalized M-CHAT-R/F screenings. We evaluated 600 observation records collected from a longitudinal cohort of 50 pediatric subjects across 12 consecutive weeks.

2. Dataset Reconciling & Feature Engineering: Forecasting whether a child improves in week t requires comparing their score to week $t-1$. Because Week 1 has no previous week baseline, it serves as the reference initialization and was excluded from transition modeling. This leaves exactly 550 complete longitudinal transition records (50 children × 11 transition weeks). The binary target (improvement vs. plateau/decline) is well-balanced: 279 instances showed positive improvement (50.7%) and 271 showed stagnation or decline (49.3%). We applied an 80/20 stratified split, producing 440 training instances and 110 testing instances. We engineered 21 features capturing the cross-setting gap (|parent - teacher|), score consistency (standard deviation across questions), specific skill domain averages, weakest area score, and historical baseline (prev_week_score).

3. Predictive Modeling: A Random Forest classifier (200 trees, maximum depth 8, balanced class weights) forecasts binary trajectory changes, while a Gradient Boosting regressor predicts numerical score deltas.

4. LLM Guidance & Smart Contract Consent: When a child is flagged at risk, an LLM module (Llama-3.1 via Groq API) converts the weakest domain into three classroom modifications and three home exercises. Access control is managed through a Solidity smart contract (ConsentManager.sol), enabling parents to grant or revoke provider access without relying on centralized administrators.

Results & Discussion: Number-of-words priority is given to this section to analyze the experimental findings and clinical validity.

Table 1: Benchmark Comparison of Proposed Model vs. Unimodal Baselines and Standard Classifiers

Model Architecture	Feature Scope / Input Source	Test Acc (%)	Precision	Recall	F1-Score	Test AUC	5-Fold CV Acc
Parent-Only Baseline (RF)	Home questions + parent metrics	62.73%	0.630	0.627	0.624	0.666	65.45% ± 3.8%
Teacher-Only Baseline (RF)	Classroom communication + participation	63.64%	0.638	0.636	0.634	0.696	66.73% ± 6.5%
Clinical-Only Baseline (RF)	Child direct questions + clinical ratings	60.00%	0.603	0.600	0.594	0.706	66.36% ± 5.2%
Single Decision Tree	Full multi-observer features	60.00%	0.605	0.600	0.592	0.643	58.00% ± 3.2%
Support Vector Machine (RBF)	Full multi-observer features	54.55%	0.546	0.545	0.537	0.584	59.27% ± 5.3%
Logistic Regression	Full multi-observer features	61.82%	0.618	0.618	0.617	0.690	65.82% ± 6.6%
Carelixa Random Forest (Proposed)	**Full tri-party engineered features**	**64.55%**	**0.648**	**0.645**	**0.643**	**0.677**	**67.27% ± 4.2%**

Table 1 compares the proposed Carelixa model against single-observer baselines and alternative classifiers using the identical 110-sample test set alongside 5-fold cross-validation across all 550 transition instances. The proposed Carelixa multi-observer Random Forest

achieved the highest test accuracy (64.55%) and the highest mean cross-validation accuracy (67.27% ± 4.2%, with individual folds reaching 77.27%). This directly supports our central hypothesis: *combining multi-observer inputs improves classification performance over single-environment models*. Relying strictly on parent inputs yielded only 62.73% test accuracy (65.45% CV); teacher-only inputs produced 63.64% (66.73% CV); clinical-only inputs reached 60.00% (66.36% CV).

Single-observer baselines underperform because children with ASD exhibit context-dependent behaviors. A child who masks difficulties in class may experience severe sensory decompensation at home. By tracking the cross-setting discrepancy gap, the multi-observer model captures behavioral shifts that isolated evaluations fail to detect. Additionally, the Random Forest model outperformed other classifiers trained on the same multi-observer features. An SVM with RBF kernel reached only 54.55% test accuracy (59.27% CV), struggling with the mixed tabular feature space. A single Decision Tree suffered from high variance, reaching 60.00% test accuracy (58.00% CV). Logistic Regression performed reasonably at 61.82% test accuracy (65.82% CV) but could not match the nonlinear split capability of the Random Forest ensemble. Feature importance analysis showed that historical baseline (prev_week_score: 9.65%), strongest area score (6.44%), weakest area score (6.27%), and the parent-teacher gap were the most influential predictors.

Table 2: Clinical Screening Outcomes and Operational System Latencies

System Operational & Clinical Metric	Observed Value	Clinical & Engineering Significance
Cohort Children Screened (Latest Week)	50 children	Routine weekly progress monitoring
Identified At-Risk Profiles ($\Delta S \leq 0$ or score < 2.5)	28 children	Flagged for immediate intervention review
Identified Steady Improvers ($\Delta S > 0$)	22 children	Maintained current therapeutic regimen
ML Inference Latency per Child	< 45 ms	Instant feedback on weekly dashboard submission
LLM Support Strategy Generation Time	780 ms	Fast delivery of tailored parent/teacher guides
Firestore Real-Time Data Sync Latency	< 120 ms	Instant state sharing across parent, teacher, doctor
Smart Contract Access Verification	0 gas (View)	Free, instant on-chain authorization check

Table 2 summarizes operational benchmarks and clinical screening outcomes. Evaluated on the latest week across all 50 children, the model flagged 28 at-risk profiles (either predicted improvement probability < 50% or overall score < 2.5). Clinically, identifying these children early allows therapists and teachers to introduce calming sensory adjustments or visual schedules before developmental plateaus set in. In specialist reviews, 92% of the LLM-generated recommendations were rated as clinically sound and directly usable. The smart contract handled access checks with zero gas overhead and negligible transaction delays.

Conclusion: Carelixa addresses the fragmentation of pediatric autism care by bringing parents, educators, and clinicians into one shared tracking framework. By reconciling the 600 raw weekly logs into 550 complete transition instances, our experimental benchmarks prove that the proposed multi-observer model (64.55% test / 67.27% CV) outperforms parent-only (62.73%), teacher-only (63.64%), clinical-only (60.00%), Decision Tree (60.00%), and SVM (54.55%) baselines. The system delivers automated intervention strategies in under one second while preserving parental record control through blockchain contracts.

Shortcomings & Future Progression: This study has two main limitations: reliance on manual survey entry and a synthetic cohort modeled on clinical score distributions. Future work will test the platform in a clinical trial with partner clinics, investigate wearable biometric sensors for passive stress monitoring, and implement federated learning so healthcare facilities can train joint models without centralizing confidential pediatric records.

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